ENV 797 - Time Series Analysis for Energy and Environment Applications | Spring 2026

Assignment 4 - Due date 02/10/26

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Directions

You should open the .rmd file corresponding to this assignment on RStudio. The file is available on our class repository on Github. And to do so you will need to fork our repository and link it to your RStudio.

Once you have the file open on your local machine the first thing you will do is rename the file such that it includes your first and last name (e.g., "LuanaLima_TSA_A04_Sp26.Rmd"). Then change "Student Name" on line 4 with your name.

Then you will start working through the assignment by **creating code and output** that answer each question. Be sure to use this assignment document. Your report should contain the answer to each question and any plots/tables you obtained (when applicable).

When you have completed the assignment, **Knit** the text and code into a single PDF file. Submit this pdf using Sakai.

R packages needed for this assignment: "xlsx" or "readxl", "ggplot2", "forecast","tseries", and "Kendall". Install these packages, if you haven't done yet. Do not forget to load them before running your script, since they are NOT default packages.\

```
#Load/install required package here
library(readxl)
library(tidyverse)
library(ggplot2)
library(forecast)
library(tseries)
library(Kendall)
```

Questions

Consider the same data you used for A3 from the spreadsheet "Table_10.1_Renewable_Energy_Production_and_Consumpti
The data comes from the US Energy Information and Administration and corresponds to the December 2025 Monthly Energy Review. **For this assignment you will work only with the column "Total Renewable Energy Production"**.

```
#Importing data set - you may copy your code from A3
filepath <- "./Data/Table_10.1_Renewable_Energy_Production_and_Consumption_by_Source.xlsx"

# load the data
energy_data_raw <- read_excel(filepath, skip = 11)
```

```r
# read in column names
colnames <- read_excel(filepath, skip = 10) %>% names()
names(energy_data_raw) <- colnames  # assign colnames to raw dataset

# basic data processing (clean names and select cols of interest)
energy_data <- energy_data_raw %>%
  janitor::clean_names() %>%
  select(month, total_renewable_energy_production)

# start date of ts data
start_date <- first(energy_data$month)
```

Stochastic Trend and Stationarity Tests

For this part you will work only with the column Total Renewable Energy Production.

Q1

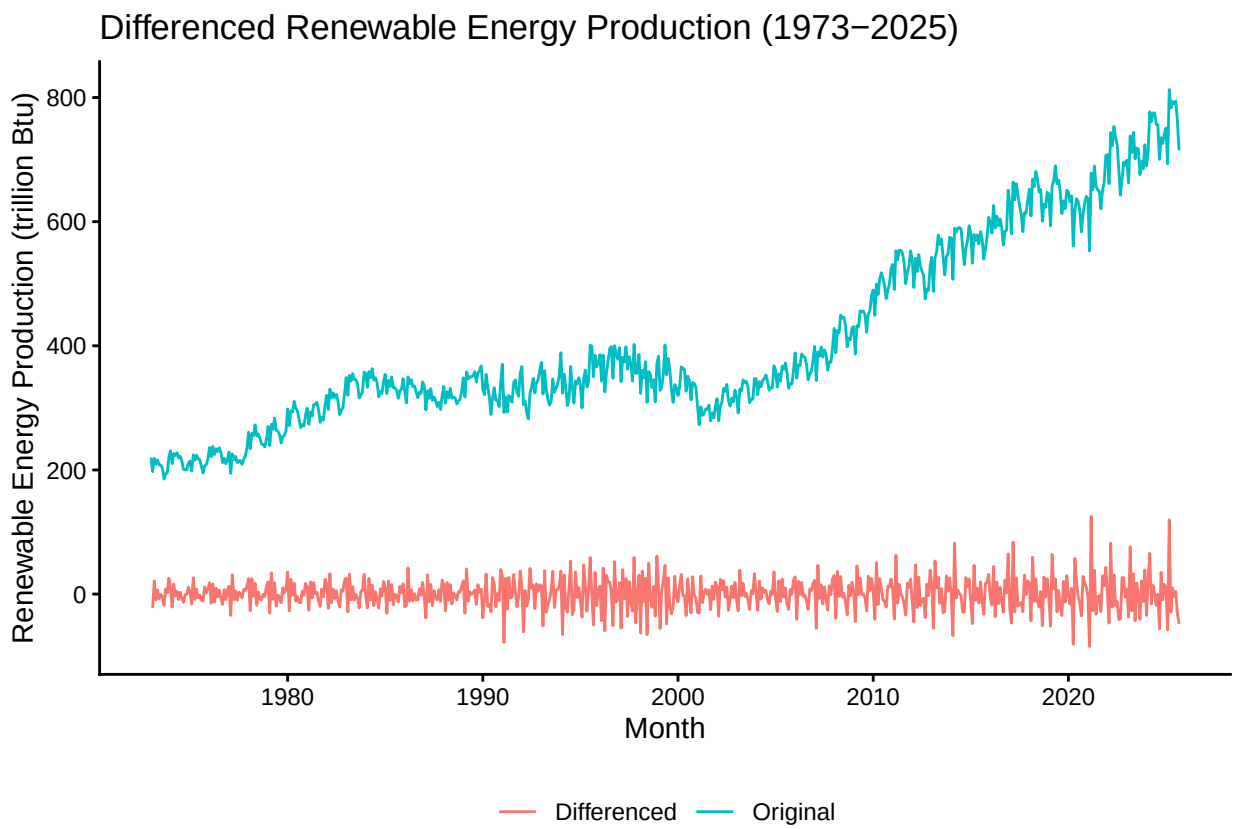
Difference the "Total Renewable Energy Production" series using function diff(). Function diff() is from package base and take three main arguments: * *x* vector containing values to be differenced; * *lag* integer indicating with lag to use; * *differences* integer indicating how many times series should be differenced.

Try differencing at lag 1 only once, i.e., make `lag=1` and `differences=1`. Plot the differenced series. Do the series still seem to have trend?

```r
renewable_series <- energy_data$total_renewable_energy_production
diff_renewable_series <- base::diff(x = renewable_series, lag = 1, differences = 1)

# convert to time series objects for plotting
ts_renewable <- ts(renewable_series,
                   start = c(year(start_date), month(start_date)),
                   freq = 12)
ts_renewable_diff <- ts(diff_renewable_series,
                        start = c(year(start_date), month(start_date)+1),
                        freq = 12)

# plot both series
autoplot(ts_renewable, series = "Original") +
  autolayer(ts_renewable_diff, series = "Differenced") +
  theme_classic() +
  theme(legend.position = "bottom") +
  labs(x = "Month",
       y = "Renewable Energy Production (trillion Btu)",
       color = NULL,
       title = "Differenced Renewable Energy Production (1973-2025)")
```



Answer: After differencing the renewable energy production time series, it no longer has a trend and all fluctuation in the series is centered around zero.

Q2

Copy and paste part of your code for A3 where you run the regression for Total Renewable Energy Production and subtract that from the original series. This should be the code for Q3 and Q4. make sure you use assign same name for the time series object that you had in A3, otherwise the code will not work.

```
t <- c(1:nrow(energy_data))  # time variable for linear model

# Total Renewable Energy Production
linear_model_renewable <- lm(renewable_series~t)

# save coefficients
renewable_b0 <- linear_model_renewable$coefficients[1] %>% as.numeric()
renewable_b1 <- linear_model_renewable$coefficients[2] %>% as.numeric()

# Subtract linear trend
renewable_series_detrend <- renewable_series - (renewable_b0 + renewable_b1*t)

# make detrended series a time series object
ts_renewable_detrend <- ts(renewable_series_detrend,
                           start = c(year(start_date), month(start_date)),
                           frequency = 12)
```

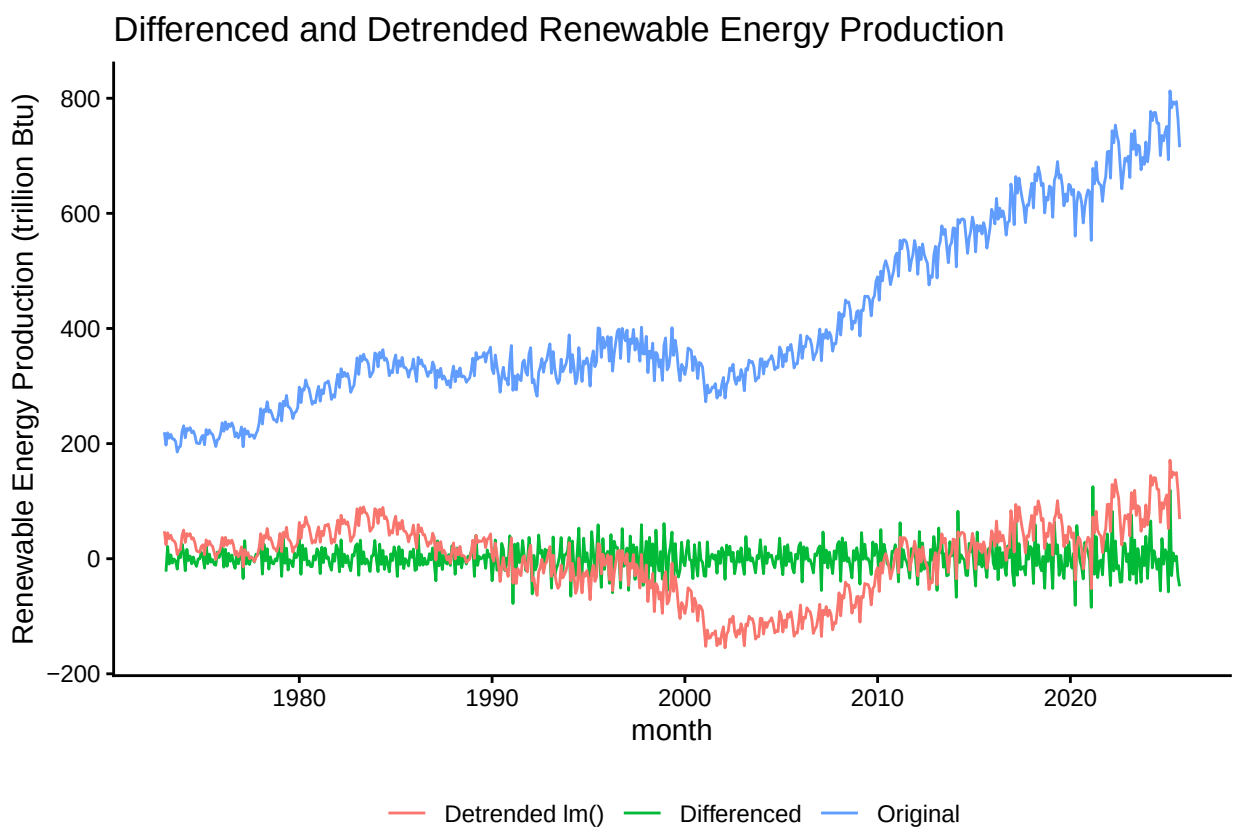
Q3

Now let's compare the differenced series with the detrended series you calculated on A3. In other words, for the "Total Renewable Energy Production" compare the differenced series from Q1 with the series you detrended in Q2 using linear regression.

Using autoplot() + autolayer() create a plot that shows the three series together (i.e. "Original", "Differenced", "Detrended lm()"). Make sure your plot has a legend. The easiest way to do it is by adding the `series=` argument to each autoplot and autolayer function. Look at the key for A03 for an example on how to use autoplot() and autolayer().

What can you tell from this plot? Which method seems to have been more efficient in removing the trend?

```
autoplot(ts_renewable, series = "Original") +
  autolayer(ts_renewable_diff, series = "Differenced") +
  autolayer(ts_renewable_detrend, series = "Detrended lm()") +
  theme_classic() +
  theme(legend.position = "bottom") +
  labs(x = "month",
       y = "Renewable Energy Production (trillion Btu)",
       title = "Differenced and Detrended Renewable Energy Production",
       color = NULL)
```



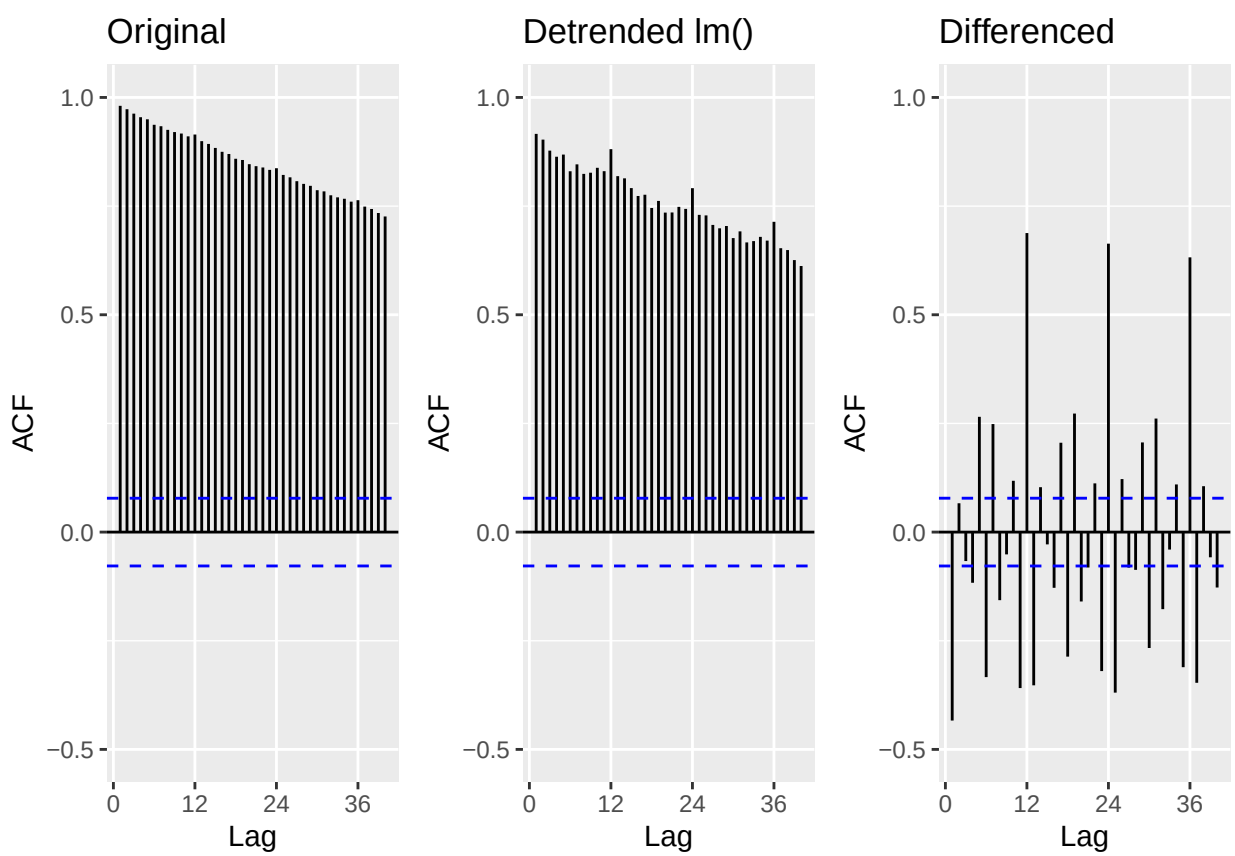
Answer: This plot shows the contrast between the differenced series and the detrended series using linear regression. While both series are centered around zero, the differenced series lacks any trend and is completely flat, while the detrended series still sees some up and down movement,

which may suggest the presence of a stochastic trend. Therefore the differencing method was more effective at removing the trend.

Q4

Plot the ACF for the three series and compare the plots. Add the argument `ylim=c(-0.5,1)` to the autoplot() or Acf() function - whichever you are using to generate the plots - to make sure all three y axis have the same limits. Looking at the ACF which method do you think was more efficient in eliminating the trend? The linear regression or differencing?

```
cowplot::plot_grid(
  ggAcf(ts_renewable, lag.max = 40, ylim=c(-0.5,1)) + labs(title = "Original"),
  ggAcf(ts_renewable_detrend, lag.max = 40, ylim=c(-0.5,1)) + labs(title = "Detrended lm()"),
  ggAcf(ts_renewable_diff, lag.max = 40, ylim=c(-0.5,1)) + labs(title = "Differenced"),
  nrow = 1
)
```



Answer: Based on the ACF plots, the differencing method was much more effective at eliminating the trend. For the detrended (lm) series, the ACF is very high for many lags and decays slowly. For the differenced series, ACF is lower or insignificant, varies between positive and negative values, and seems to spike at or around the seasonal lags (i.e. every 12 months).

Q5

Compute the Seasonal Mann-Kendall and ADF Test for the original "Total Renewable Energy Production" series. Ask R to print the results. Interpret the results for both test. What is the conclusion from the Seasonal Mann Kendall test? What's the conclusion for the ADF test? Do they match what you observed in Q3 plot? Recall that having a unit root means the series has a stochastic trend. And when a series has stochastic trend we need to use differencing to remove the trend.

```
smk_original <- Kendall::SeasonalMannKendall(ts_renewable)
summary(smk_original)
```

```
## Score =  13083 , Var(Score) = 201135
## denominator =  16379.5
## tau = 0.799, 2-sided pvalue =< 2.22e-16
```

```
adf_original <- adf.test(ts_renewable)
adf_original
```

```
##
##  Augmented Dickey-Fuller Test
##
## data:  ts_renewable
## Dickey-Fuller = -1.0247, Lag order = 8, p-value = 0.9347
## alternative hypothesis: stationary
```

> Answer: Based on the Seasonal Mann-Kendall test of the original data, we can reject the null hypothesis (p-value $<$ 2.22e-16) that the series is stationary, meaning that it does contain a deterministic trend. Based on the ADF of the original series, we accept the null hypothesis (p-value = 0.9347) that the series is not stationary and contains a unit root. The ADF test confirms my observation in the Q3 plot that there was a trend present even after removing the linear trend component, which suggested the series has a stochastic trend.

Q6

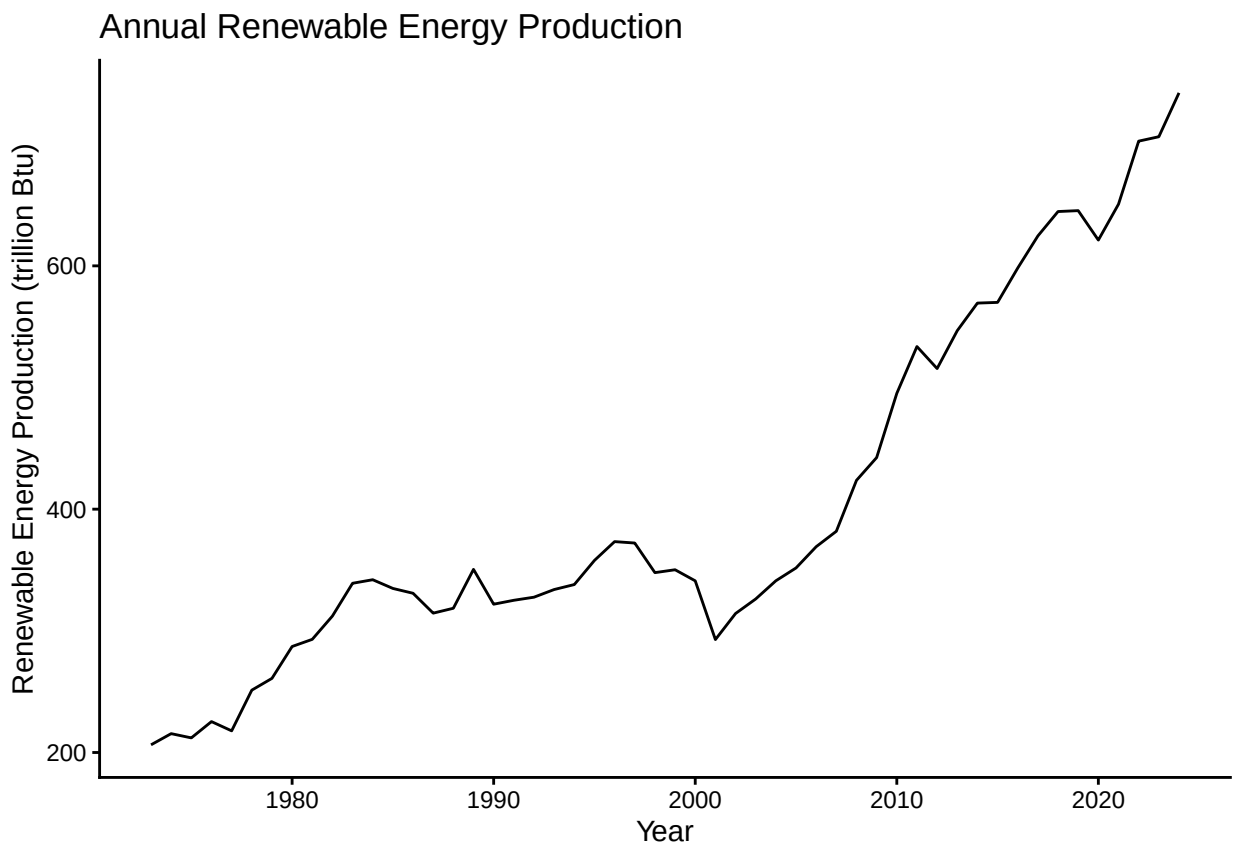
Aggregate the original "Total Renewable Energy Production" series by year. You can use the same procedure we used in class. Store series in a matrix where rows represent months and columns represent years. And then take the columns mean using function colMeans(). Recall the goal is the remove the seasonal variation from the series to check for trend. Convert the accumulates yearly series into a time series object and plot the series using autoplot().

```
# Exclude the last year of data (2025) from the series because it is incomplete
renewable_series_annual <- matrix(renewable_series[1:(floor(length(renewable_series)/12)*12)],
                                  nrow = 12, byrow = FALSE) %>%
  colMeans()

# convert to time series object
ts_renewable_annual <- ts(renewable_series_annual, start = year(start_date), frequency = 1)

# plot annual time series
autoplot(ts_renewable_annual) +
  theme_classic() +
```

```r
  labs(x = "Year",
       y = "Renewable Energy Production (trillion Btu)",
       title = "Annual Renewable Energy Production")
```



Q7

Apply the Mann Kendall, Spearman correlation rank test and ADF. Are the results from the test in agreement with the test results for the monthly series, i.e., results for Q5?

```r
# Mann-Kendall Test
mk_annual <- Kendall::MannKendall(ts_renewable_annual)
summary(mk_annual)
```

```
## Score =  1084 , Var(Score) = 16059.33
## denominator =  1326
## tau = 0.817, 2-sided pvalue =< 2.22e-16
```

```r
# Spearman Correlation Rank Test
t <- c(year(start_date):(year(start_date)+length(ts_renewable_annual)-1))
spearman_corr_annual <- cor.test(x = t, y = ts_renewable_annual, method = "spearman")
spearman_corr_annual
```

```
##
```

```
##  Spearman's rank correlation rho
##
## data:  t and ts_renewable_annual
## S = 1852, p-value < 2.2e-16
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
##       rho
## 0.9209425
```

```r
# Augmented Dickey Fuller Test
adf_annual <- adf.test(ts_renewable_annual)
adf_annual
```

```
##
##  Augmented Dickey-Fuller Test
##
## data:  ts_renewable_annual
## Dickey-Fuller = -0.85301, Lag order = 3, p-value = 0.9515
## alternative hypothesis: stationary
```

Answer: The Mann-Kendall test of the annualized data confirms that the data is not stationary and has a deterministic trend (p-value < 2.22e-16). The Spearman correlation rank test found a that there is a positive monotonic trend (rho = 0.921, p-value < 2.22e-16), which aligns with our previous observations. The ADF test of the annualized data found that the data is not stationary and contains a unit root (p-value = 0.9515). This confirms our conclusion that even after removing the seasonal component, the series shows a stochastic trend.